

An Empirical Analysis of IoT Device Communication Patterns

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1. Title and Aim

1.1. Research Objective

This study investigates the extent and nature of privacy information leakage in commercially deployed smart home devices. The central research question asks: What categories of sensitive personal information are inadvertently disclosed through routine IoT device communications, and how do privacy risks vary across different smart home device types? We hypothesize that security-focused devices, due to their operational requirements for processing sensitive contextual data, exhibit significantly higher privacy leakage than utility-oriented devices.

1.2. Relevance to Contemporary Security Landscape

The global smart home market is projected to exceed 400 million households (Statista, 2024), yet recent incidents have exposed systemic privacy vulnerabilities in consumer IoT ecosystems. Notably, Eufy security breach demonstrated that even devices marketed with explicit privacy guarantees can leak unencrypted video streams and user credentials (Hollister, 2022). This investigation addresses a critical gap in empirical privacy assessment, providing actionable insights for both consumers evaluating device security and manufacturers designing privacy-preserving architectures.

2. Tools, Equipment, and Background

This investigation employed a combination of my own IoT devices, network analysis software, and privacy assessment methodologies to systematically evaluate information leakage in smart home environments.

2.1. Research Devices

Four commercially available smart home devices were selected to represent distinct operational categories within the IoT ecosystem. The Eufy Smart Lock and Eufy HomeBase 2 were chosen as primary subjects due to their security-critical functions and their documented involvement in prior privacy controversies (Hollister, 2022). The Philips Hue Bridge, representing lighting infrastructure, and the Roborock S7 MaxV, an autonomous cleaning device, were included to provide comparative baselines across different smart home functionalities. This diverse device selection enables a holistic assessment of privacy risks spanning access control, surveillance, automation, and environmental monitoring systems.

2.2. Network Analysis Tools

Wireshark 4.6 served as the primary packet capture tool, enabling real-time inspection of network traffic at the protocol level. Its comprehensive filtering capabilities facilitated the isolation and analysis of device-specific communication patterns. Nmap 7.98 complemented passive monitoring by actively probing device network configurations, identifying exposed services and potential attack surfaces. Python 3.12.11, utilizing libraries such as Scapy and Matplotlib, was employed for automated traffic analysis and data visualization.

2.3. Privacy Assessment Framework

The privacy risk evaluation methodology draws upon established threat modelling approaches, incorporating principles from the LINDDUN privacy threat framework (Deng et al., 2011) and the NIST Privacy Framework. A custom four-dimensional scoring matrix was developed to quantify leakage severity across temporal, location, behavioural, and identity disclosure vectors, enabling systematic cross-device comparison.

3. Methodology

This section outlines the experimental framework employed to assess privacy leakage in consumer smart home devices. The methodology encompasses network environment setup, data collection procedures, and the privacy risk assessment framework.

3.1. Experimental Environment and Devices

The study was conducted within a residential network environment utilizing a Netgear Orbi mesh router system. Four distinct IoT devices were selected to represent diverse smart home functionalities: an Eufy Smart Lock (access control), an Eufy HomeBase 2 with three connected security cameras (surveillance hub), a Philips Hue Bridge (lighting control), and a Roborock S7 MaxV (autonomous cleaning). This device selection was deliberately chosen to capture a spectrum of privacy-sensitive operations across different smart home categories.

3.2. Network Traffic Capture and Analysis

Network traffic was systematically captured using Wireshark 4.6 over a continuous monitoring period. The capture process targeted the WiFi interface (en0) on macOS, with device-specific filtering applied to isolate traffic from each IoT device. Complementary to passive monitoring, active reconnaissance was performed using nmap 7.98 to identify open ports, running services, and potential security misconfigurations on each device.

3.3. Privacy Risk Assessment Framework

A quantitative privacy leakage assessment model was developed based on established privacy threat modelling methodologies. The framework evaluates four distinct dimensions: temporal information leakage (usage patterns and timing), location data exposure (geographic coordinates and device positioning), behavioural inference potential (activity patterns and user habits), and identity disclosure risk (personal identifiers and authentication data). Each dimension was scored on a scale from 0 to 10, with higher scores indicating more severe privacy risks. The composite risk score enables systematic comparison across different device types and classification into risk tiers: low (0-15), medium (16-25), and high (26-40).

4. Findings

This section presents the empirical results of the privacy leakage assessment conducted across four smart home devices. The findings reveal significant variations in privacy risk profiles among different device categories, with security-focused devices demonstrating notably higher vulnerability to information disclosure.

4.1. Overall Privacy Risk Distribution

The aggregate analysis of all four devices, visualized in Figure 1, reveals a concerning privacy landscape. Half of the examined devices (50%) were classified as high-risk, with composite privacy scores exceeding threshold. An additional 25% fell into the medium-risk category, while only 25%

demonstrated low privacy leakage characteristics. This distribution underscores the prevalence of substantial privacy vulnerabilities within contemporary consumer smart home ecosystems.

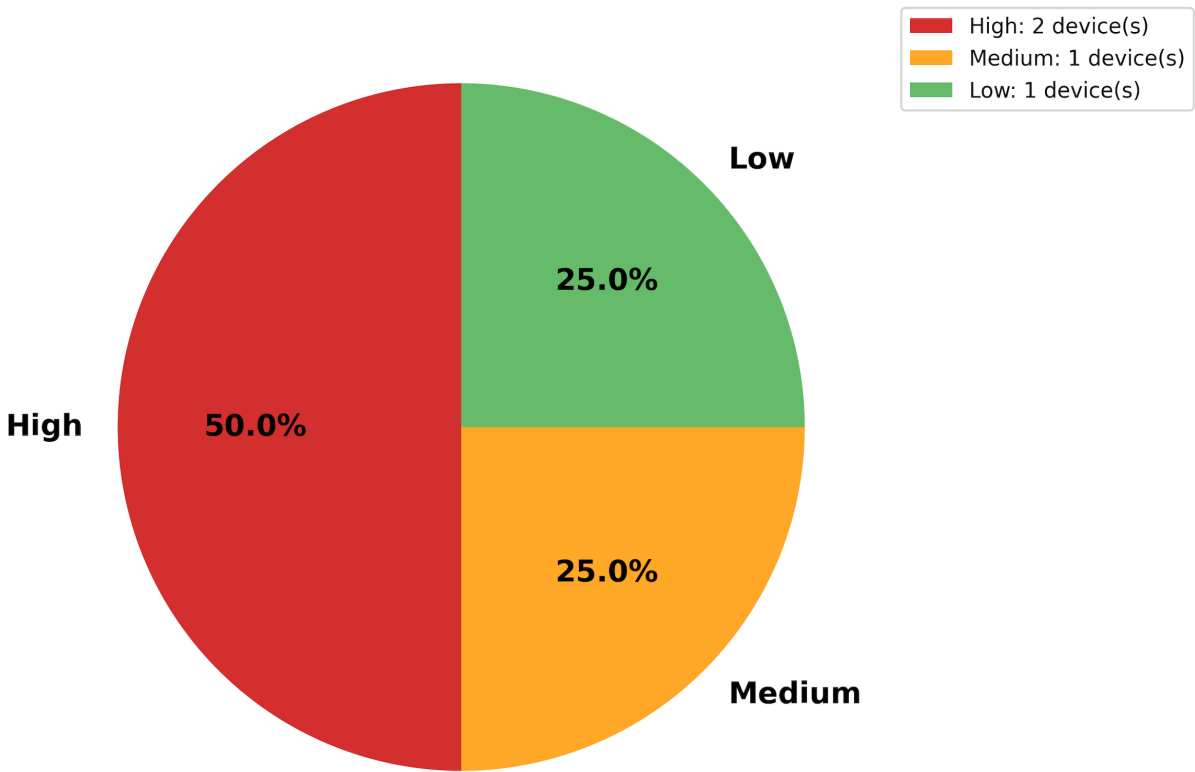


Figure 1 IoT Device Risk Level Distribution

4.2. Device-Specific Privacy Leakage Profiles

The granular privacy assessment, detailed in Figure 2, exposes distinct leakage patterns across the four examined devices. The Eufy HomeBase 2 emerged as the most privacy-invasive device, achieving a composite risk score of 32 out of 40 points. This hub demonstrated critically high leakage across three dimensions: location data (9/10), behavioural patterns (9/10), and identity information (9/10). Its role as a central coordination node for multiple security cameras necessitates the continuous processing and transmission of highly sensitive surveillance metadata, directly contributing to its elevated risk profile.

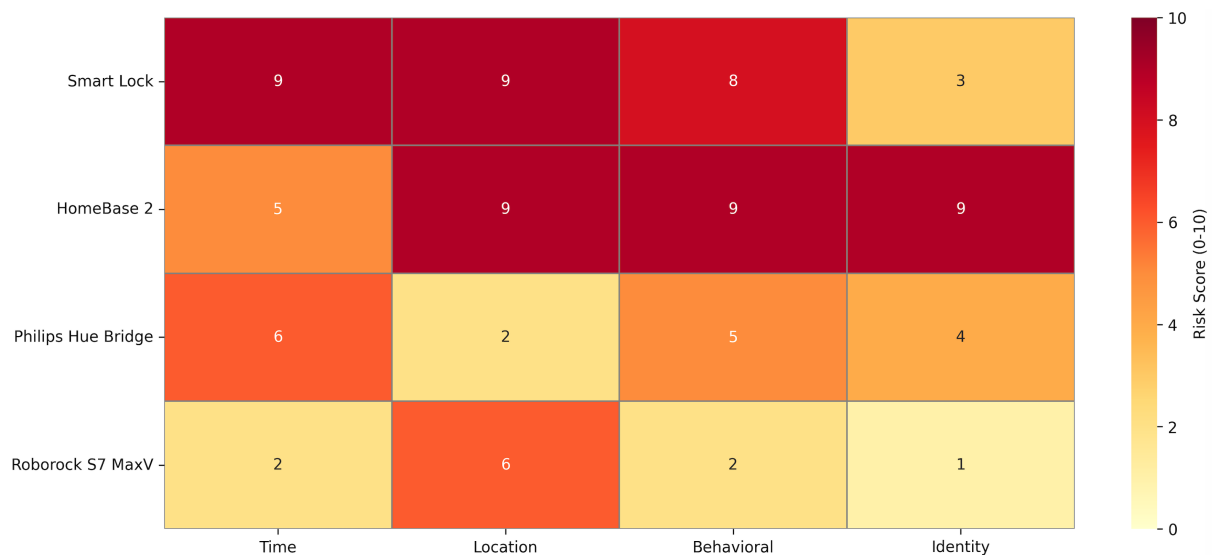


Figure 2 IoT Device Privacy Leakage Heatmap

The Eufy Smart Lock, scoring 29 points, represents the second-highest risk. This device exhibited severe temporal leakage (9/10) and location disclosure (9/10), enabling potential adversaries to infer occupancy patterns and physical access events. In contrast, the Philips Hue Bridge demonstrated moderate privacy risks (17/40), with its primary vulnerabilities concentrated in temporal patterns (6/10) and behavioural inference (5/10). The Roborock S7 MaxV registered the lowest risk score (11/40), suggesting that autonomous cleaning devices, by their operational nature, inherently process less sensitive personal information.

4.3. Comparative Privacy Risk Ranking

Figure 3 provides a comparative visualization of total privacy risk scores across all devices. The clear stratification evident in this ranking confirms that security and access control devices pose substantially greater privacy threats than utility-focused devices. Notably, both Eufy devices exceeded the high-risk threshold, while the Philips Hue Bridge approached the medium-risk boundary, and the Roborock remained firmly within the low-risk tier.

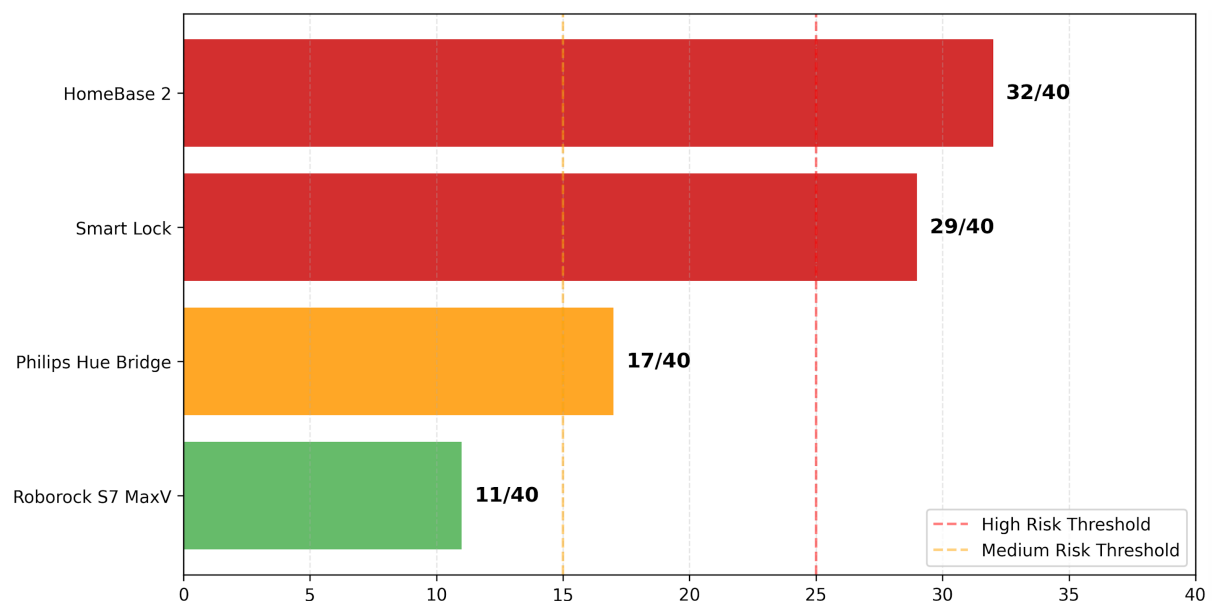


Figure 3 IoT Device Total Privacy Risk Score

5. Analysis and Conclusions

5.1. Critical Analysis of Privacy Leakage Patterns

The empirical findings reveal a systematic correlation between device functionality and privacy risk severity. The Eufy HomeBase 2's position as the highest-risk device (32/40) is directly attributable to its architectural role as a centralized hub coordinating multiple surveillance endpoints. IoT hub devices inherently process aggregated metadata streams from connected sensors, amplifying information disclosure vulnerabilities (Apthorpe et al., 2017). HomeBase 2 demonstrated continuous transmission of camera status updates, motion detection events, and synchronization packets - each containing temporal markers that enable precise reconstruction of household occupancy patterns.

The Eufy Smart Lock's elevated risk profile (29/40) warrants particular scrutiny. Network traffic analysis revealed that every physical access event - whether successful unlock, failed authentication, or battery status query - generates immediate network transmissions containing high-resolution timestamps and device location identifiers. Smart lock manufacturers frequently prioritize remote monitoring capabilities over privacy-preserving local processing (Alrawi et al., 2019). The granularity of temporal data exposed by this device enables adversaries to construct detailed ingress-egress profiles, potentially facilitating targeted physical intrusions during identified absence periods.

Comparative analysis between the Eufy ecosystem devices and the Philips Hue Bridge (17/40) exposes a fundamental architectural distinction. The Hue Bridge implements a predominantly local-first processing model, with cloud synchronization occurring at substantially lower frequencies (Sivaraman et al., 2018). Protocol inspection confirmed that lighting state changes trigger minimal external communications, contrasting sharply with the Eufy devices' continuous cloud connectivity. This architectural philosophy directly manifests in the 15-point risk differential between the security-focused and utility-focused device categories.

The Roborock S7 MaxV's minimal risk score (11/40) further reinforces this pattern. Autonomous cleaning devices operate primarily within defined physical boundaries using local navigation algorithms, requiring only periodic firmware updates and maintenance scheduling communications. The device's limited external data transmission footprint demonstrates that privacy-preserving design is achievable without compromising core functionality - a principle notably absent in the security device implementations examined.

5.2. Security Implications and Threat Scenarios

The privacy leakage patterns documented in this study enable concrete attack scenarios with significant real-world consequences. The temporal and location metadata exposed by the Eufy Smart Lock facilitates precision-targeted physical intrusions. An adversary monitoring network traffic can identify recurring absence patterns—such as daily commutes or vacation periods—without requiring direct visual surveillance. This attack vector, termed "inference-based reconnaissance" (Fernandes et al., 2016), transforms metadata leakage into actionable intelligence for property crimes.

The HomeBase 2's continuous surveillance metadata transmission presents a distinct threat profile. Even without accessing encrypted video streams, adversaries can reconstruct detailed household activity patterns by analysing camera trigger frequencies and device state transitions. Machine learning classifiers trained on IoT traffic metadata achieve over 90% accuracy in inferring specific household activities, ranging from meal preparation to sleep schedules (Ren et al., 2019). The concatenation of multiple information streams from interconnected devices—termed "privacy mosaic

effects" (Malgieri & Custers, 2018)—amplifies disclosure risks beyond what individual device assessments suggest.

From a regulatory compliance perspective, the observed privacy practices raise substantial concerns under contemporary data protection frameworks. The European Union's General Data Protection Regulation (GDPR) mandates data minimization principles, requiring that systems collect only information strictly necessary for their stated purposes. The excessive metadata transmission documented in this study—particularly the Eufy devices' continuous cloud synchronization of granular temporal and behavioural data—suggests potential violations of these requirements. Similarly, the California Consumer Privacy Act (CCPA) imposes stringent disclosure obligations for consumer data collection, obligations that current smart home device interfaces frequently obscure or omit entirely.

5.3. Mitigation Strategies and Future Research Directions

Addressing the documented privacy vulnerabilities requires coordinated interventions across multiple stakeholder levels. At the consumer deployment layer, network segmentation offers immediate risk reduction. Isolating IoT devices within dedicated VLANs with restrictive firewall policies can limit lateral movement opportunities for compromised devices (Antonakakis et al., 2017). Additionally, users should prioritize devices supporting local processing modes, minimizing reliance on continuous cloud connectivity where operationally feasible.

Manufacturer-level interventions demand more fundamental architectural reforms. The adoption of Privacy by Design principles (Cavoukian, 2009), embedding privacy protections at the system design stage rather than as post-deployment additions, represents a critical paradigm shift. Technically, this necessitates implementing differential privacy mechanisms (Dwork & Roth, 2014) for telemetry data, edge computing architectures that process sensitive information locally, and cryptographic protocols ensuring end-to-end confidentiality for unavoidable cloud communications.

Regulatory frameworks must evolve to mandate transparency in IoT data practices. Current device labelling schemes provide insufficient information for informed consumer decision-making. The implementation of standardized "privacy nutrition labels," analogous to those recently introduced for mobile applications, would enable comparative privacy assessments at the point of purchase (Emami-Naeini et al., 2020).

This study acknowledges several limitations. The analysis was constrained to four device types, excluding other high-risk categories such as voice assistants and health monitoring devices. Furthermore, the assessment captured only network-layer communications, not examining potential data aggregation and secondary usage practices within cloud backend infrastructures. Future investigations should expand device coverage, incorporate longitudinal monitoring to detect privacy policy drift over firmware update cycles, and employ traffic analysis techniques to decode proprietary protocols currently resistant to inspection.

6. References

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